

An Observation on the Isolated Peak Wave of a Constant-Amplitude Odd-Harmonic Sum on a Half-Wavelength Phase Interval and Its Localization

Subtitle: Taking the fundamental domain to be the half-wavelength $[-\pi/2, \pi/2]$, the constant-amplitude odd-harmonic sum becomes an isolated peak wave with a peak at the center, and the localization width of its squared amplitude shrinks as $1/(N+1)$ (the reciprocal of the highest odd-harmonic order N plus one; for large N , essentially the reciprocal of the highest harmonic order). A formula for the required highest harmonic order, inverted from a prescribed localization width, is also given.

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Position: First draft as an observational and organizing paper. It records, as an elementary property of Fourier sums, that superposing constant-amplitude odd harmonics on a half-wavelength phase interval produces a waveform with a peak at the center (here called an “isolated peak wave”). It does not derive physical laws, assert observational facts, or give any particular physical interpretation.

Abstract

This paper observes, for a wave formed by superposing constant-amplitude odd harmonics on the half-wavelength phase interval $\varphi \in [-\pi/2, \pi/2]$, that it becomes an isolated peak wave with a peak at the center, together with the relation between its localization width and the highest harmonic order. The wave under consideration is defined by

$$S_N(\varphi) = \sum_{m=0}^{(N-1)/2} \cos((2m+1)\varphi)$$

where N is the highest odd-harmonic order (N odd), and the sum contains the $(N+1)/2$ terms $1, 3, 5, \dots, N$.

This paper records this property only as a mathematical observation of an odd-harmonic superposition on a half-wavelength phase interval, and gives no physical interpretation.

1. Introduction

In Fourier analysis it is widely known that introducing higher harmonics shapes the local structure of a waveform. Here we restrict attention to odd harmonics with constant amplitude and observe the single isolated peak wave they form at the center of the half-wavelength interval, together with its localization.

The wave treated here, a constant-amplitude superposition of odd harmonics only, (a) uses cosine $\cos((2m+1)\varphi)$

rather than sine so that the center is a peak, and (b) keeps the coefficients at the constant amplitude 1. In this paper we shall, for convenience, call a waveform that in this way has a dominant main peak at the center and vanishes at both ends of the half-wavelength interval an “isolated peak wave.”

Notation and formulas follow a standard analysis text [1].

Note that the closed form of Eq. (2.3) below can also be written as a special case of the Dirichlet kernel; however, the aim of this paper is not to discuss the theoretical properties of the Dirichlet kernel, but to start directly from the constant-amplitude odd-harmonic sum on a half-wavelength phase interval and to organize its localization shape, the scaling of its localization width, and the inverse formula for the required harmonic order. We therefore proceed below without presupposing the Dirichlet kernel, deriving the wave under consideration independently.

2. Definitions and Fundamental Domain

2.1 Definition of the wave

For the variable $\varphi \in [-\pi/2, \pi/2]$, we define the wave amplitude $S_N(\varphi)$ by

$$S_N(\varphi) = \sum_{m=0}^{(N-1)/2} \cos((2m+1)\varphi) \quad (2.1)$$

where N is the highest odd-harmonic order and N is odd.

Because the amplitude $S_N(\varphi)$ takes both positive and negative values, we use as the non-negative observed quantity the squared amplitude

$$I_N(\varphi) = |S_N(\varphi)|^2 = S_N(\varphi)^2 \quad (2.2)$$

Figure 1 shows this squared amplitude I_N , normalized to a maximum of 1 on the half-wavelength interval (for $N = 99, 999, 9999$). The shape of the isolated peak wave, with a main peak at the center and zeros at both ends, is visible, and the higher the highest harmonic order N , the narrower the central main peak.

Figure 1: Isolated peak wave (the normalized squared amplitude of the constant-amplitude odd-harmonic sum, $N = 99, 999, 9999$).

2.2 Cosine sum

From the cosine-sum formula for the arithmetic progression $a + md$ ($a = \varphi$, $d = 2\varphi$) [1],

$$S_N(\varphi) = \sum_{m=0}^{(N-1)/2} \cos((2m+1)\varphi) = \frac{\sin((N+1)\varphi)}{2 \sin \varphi} \quad (2.3)$$

This Eq. (2.3) is an exact identity used repeatedly in what follows (at $\varphi = 0$ the right-hand side is read as $\lim_{\varphi \rightarrow 0}$, giving $S_N(0) = (N+1)/2$ stated below).

Figure 1. Isolated peak wave: the equal-amplitude odd-harmonic sum on the half-wavelength interval

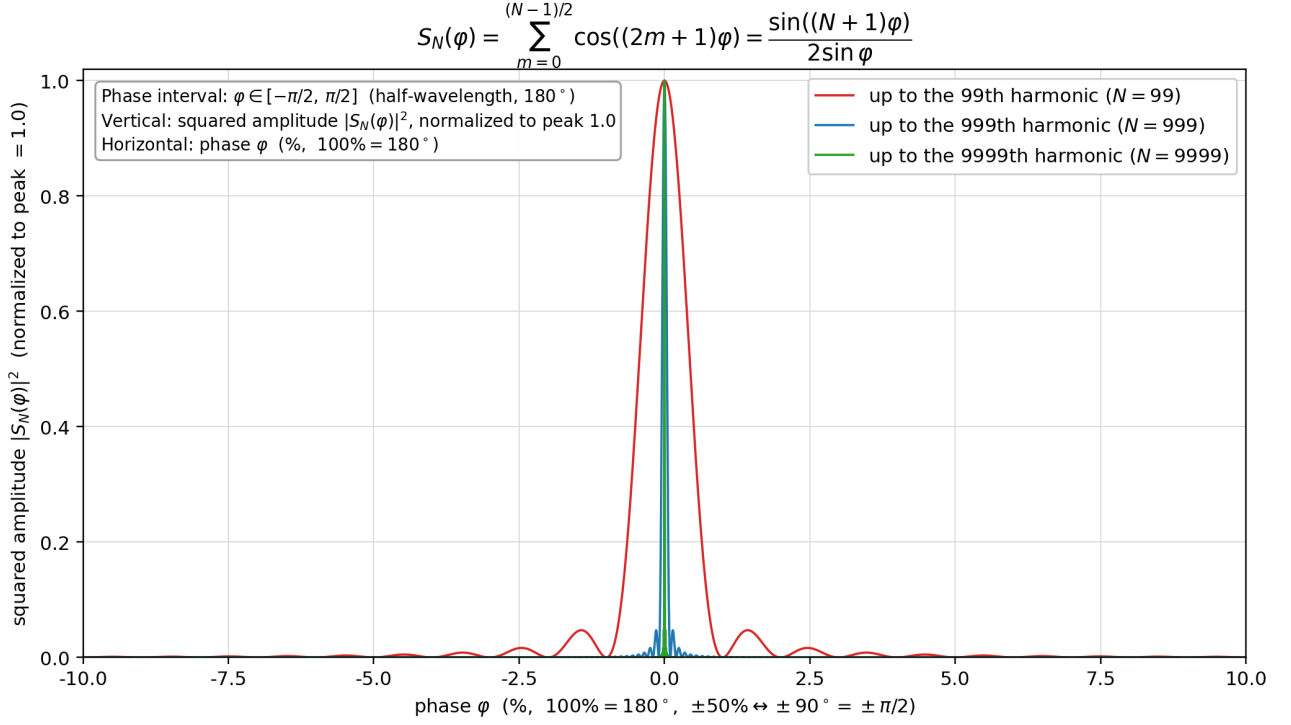


Figure 1

2.3 The $\sin u/u$ approximation near the central main peak

Here we derive, step by step from Eq. (2.3), the approximate form used in the later discussion of the localization width (§2.4). The goal is to show that “if we fix the magnified variable $u = (N+1)\varphi$ and look near the central main peak, then the shape of S_N normalized by its peak value approaches the universal function $\sin u/u$, independent of N .”

Step 1 (magnified variable near the peak). To observe only the neighborhood of the central main peak ($\varphi = 0$), introduce the variable stretched to the peak width,

$$u = (N+1)\varphi, \quad \text{that is, } \varphi = \frac{u}{N+1} \quad (2.4)$$

This is a magnifying lens that keeps the interior of the peak at a fixed resolution however large N becomes; the range $u = O(1)$ corresponds to the neighborhood of the central main peak. Substituting into Eq. (2.3), the numerator’s argument $(N+1)\varphi$ becomes exactly u , so

$$S_N(\varphi) = \frac{\sin((N+1)\varphi)}{2\sin\varphi} = \frac{\sin u}{2\sin(u/(N+1))} \quad (2.5)$$

This is still exact; no approximation has entered. The denominator $2\sin\varphi$ has not vanished but been rewritten as $2\sin(u/(N+1))$.

Step 2 (small-angle expansion of the denominator). Keeping $u = O(1)$ (near the central main peak) and letting $N \rightarrow \infty$, the denominator’s argument $u/(N+1)$ becomes arbitrarily small. Applying the small-angle expansion $\sin\theta = \theta + O(\theta^3)$ with $\theta = u/(N+1)$,

$$\sin\left(\frac{u}{N+1}\right) = \frac{u}{N+1} + O\left(\frac{u^3}{(N+1)^3}\right) \quad (2.6)$$

For $u = O(1)$ (near the central main peak) the error is $O((N+1)^{-3})$. Substituting into the denominator of Eq. (2.5),

$$S_N(\varphi) \approx \frac{\sin u}{2 \frac{u}{N+1}} = \frac{N+1}{2} \cdot \frac{\sin u}{u} \quad (2.7)$$

All N -dependence has collected into the prefactor $\frac{N+1}{2}$, and the u -dependent part has separated into the universal form $\sin u/u$, independent of N .

Step 3 (normalization by the peak value). Since $\sin u/u \rightarrow 1$ as $u \rightarrow 0$, the peak value is

$$S_N(0) = \frac{N+1}{2} \quad (2.8)$$

(which also follows directly from each cosine in Eq. (2.1) being 1 at $\varphi = 0$, with $(N+1)/2$ terms). To compare only the shape, not the actual height of the wave, define the **normalized amplitude** and **normalized squared amplitude**, divided by the peak value, by

$$\widehat{S}_N(\varphi) := \frac{S_N(\varphi)}{S_N(0)}, \quad \widehat{I}_N(\varphi) := \frac{I_N(\varphi)}{I_N(0)} = \widehat{S}_N(\varphi)^2$$

(the vertical-axis normalization in Figures 1 and 2, and the $\widehat{I}_N$ used from Eq. (2.10) on, refer to this quantity). Dividing Eq. (2.7) by Eq. (2.8), the prefactor $\frac{N+1}{2}$ cancels exactly, giving

$$\widehat{S}_N(\varphi) \approx \frac{\sin u}{u}, \quad \widehat{I}_N(\varphi) = \widehat{S}_N(\varphi)^2 \approx \left(\frac{\sin u}{u}\right)^2 \quad (u = (N+1)\varphi) \quad (2.9)$$

This is the approximate form used from Eq. (2.10) on. The key points are three: (i) the magnified variable $u = (N+1)\varphi$ views the central main peak at fixed resolution; (ii) under it the denominator $2 \sin \varphi$ turns into $2u/(N+1)$; and (iii) normalizing by the peak value $\frac{N+1}{2}$ makes the N -dependence drop out completely. Moreover, that the pre-normalization localization width is $\varphi \sim u/(N+1)$, i.e. shrinks as $1/(N+1)$, can be read off from Eq. (2.4).

Figure 2 shows ($N = 99, 999, 9999$) together. The horizontal axis is varied by factors of 10, $\pm 10\%$, $\pm 1\%$, $\pm 0.1\%$. The vertical axis is shared across the three panels, with $|S_N(\varphi)|^2$ normalized to a maximum of 1.0. The figure is to confirm visually that the width of the central main peak varies on the $1/(N+1)$ horizontal scale.

Figure 2: $1/(N+1)$ scaling of the localization width.

2.4 Required highest harmonic order $N_{\min}(\Delta_k, k)$ from a prescribed localization width

We show how to solve “how high must the highest odd-harmonic order N be taken to squeeze the central main peak to a prescribed thinness.”

Use the normalized phase $x = \varphi/\pi \in [-\frac{1}{2}, \frac{1}{2}]$, taking the full width π of the half-wavelength interval as 1. For a tolerance level k ($0 < k < 1$), define the **k -localization half-width Δ_k** as the positive position at which the normalized squared amplitude first drops to k or below,

Figure 2. The same three curves ($N = 99$ red, 999 blue, 9999 green) overlaid at three zooms ($\pm 10\%$, $\pm 1\%$, $\pm 0.1\%$). In each window exactly one main lobe fits; the others are $\times 10$ wider or narrower, so the localization width scales as $W(N) \propto 1/N$.

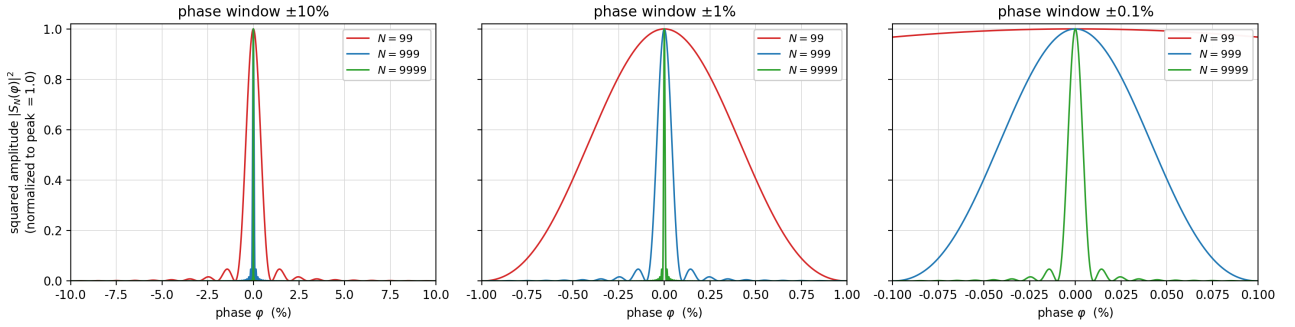


图 2 Figure 2

$$\Delta_k = \min\{x > 0 : \hat{I}_N(\pi x) \leq k\} \quad (2.10)$$

For large N , by Eq. (2.9) of §2.3,

$$\hat{I}_N(\varphi) \approx (\sin u/u)^2 \quad (u = (N+1)\varphi)$$

so $\hat{I}_N$ first drops to k when u reaches the first positive root u_k of

$$\left(\frac{\sin u_k}{u_k}\right)^2 = k \quad (2.11)$$

with $0 < u_k < \pi$, i.e. at $\varphi = u_k/(N+1)$.

Here u_k is a **specific value** of $u = (N+1)\varphi$, namely the point at which the left-hand side of Eq. (2.11) drops to the level k , and is a constant determined by the tolerance level k alone. Since $g(u) = \sin u/u$ decreases monotonically from 1 to 0 on $[0, \pi]$, it has an inverse, and u_k is given uniquely by that inverse:

$$u_k = g^{-1}(\sqrt{k}), \quad g(u) = \frac{\sin u}{u} \quad (0 \leq u \leq \pi). \quad (2.12)$$

However, Eq. (2.11) is a transcendental equation, and u_k cannot be written in closed form with elementary functions. In practice one uses the numerical solution (table below), but for small k the approximate closed form

$$u_k \approx \frac{\pi}{1 + \sqrt{k}} \quad (2.13)$$

holds (a first-order approximation setting $u_k = \pi - \delta$ and $\sin \delta \approx \delta$; the relative error is below 0.15% for $k \leq 0.01$ and below 0.01% for $k \leq 0.001$. By construction the accuracy worsens as k grows: about 3% at $k = 0.1$ and about 30% at $k = 0.5$).

From the condition $u_k/(N+1) \leq \pi \Delta_k$ that the localization half-width be within Δ_k ,

$$N \geq \frac{u_k}{\pi \Delta_k} - 1 \quad (2.14)$$

We distinguish two inversion methods. First, the method that uses the u_k obtained by solving Eq. (2.11) numerically is called here the **numerical-substitution method**. It is defined by the single expression

$$\begin{aligned}
N_{\text{cont}}^{(\text{num})}(\Delta_k, k) &:= \frac{u_k}{\pi \Delta_k} - 1, \quad \left(\frac{\sin u_k}{u_k} \right)^2 = k, \quad 0 < u_k < \pi, \\
N_{\text{min}}^{(\text{num})}(\Delta_k, k) &:= \min \left\{ N \in 2\mathbb{Z}_{\geq 0} + 1 : N \geq N_{\text{cont}}^{(\text{num})}(\Delta_k, k) \right\}.
\end{aligned} \tag{2.15}$$

Here $N_{\text{cont}}^{(\text{num})}$ is the continuous threshold before imposing the odd-integer condition, and $N_{\text{min}}^{(\text{num})}$ is the smallest odd integer at least as large. This inversion formula uses the numerical solution of Eq. (2.11) directly, without replacing u_k by elementary functions.

As a comparison example, substituting the approximation (2.13) to eliminate u_k yields the **approximate closed form**

$$N_{\text{cont}}^{(\text{app})}(\Delta_k, k) := \frac{1}{(1 + \sqrt{k}) \Delta_k} - 1 \tag{2.16}$$

(using $u_k \approx \pi/(1 + \sqrt{k})$; Δ_k is the normalized half-width with full width π taken as 1). However, Eq. (2.16) is an estimate stemming from the approximation of u_k , and since the numerical-substitution method (2.15) is fast enough in practice, there is no advantage to using the less accurate (2.16).

Adding $N = 99999, 999999$ to the $N = 99, 999, 9999$ used in Figures 1 and 2, and taking the half-width $\Delta_{0.01}$ at $k = 0.01$ as input, the numerical-substitution method (2.15) and the approximate closed form (2.16) compare as follows:

$\Delta_{0.01}$ (closed-form				
Original highest odd-harmonic order	N	$\Delta_{0.01}$ (evaluation equivalent to the direct sum, percent)	$N_{\text{cont}}^{(\text{num})}$ (Eq. (2.15))	$N_{\text{min}}^{(\text{num})}$ (Eq. (2.15))
				$N_{\text{cont}}^{(\text{app})}$ (Eq. (2.16))
	99	0.907940255%	98.998719	99
	999	0.0907928740%	998.999872	999
	9999	0.00907928625%	9998.999987	9999
	99999	0.000907928624%	99998.999999	99999
	999999	0.0000907928624%	999998.999999	999999

3. Conclusion

For the constant-amplitude odd-harmonic sum on the half-wavelength phase interval $\varphi \in [-\pi/2, \pi/2]$,

$$S_N(\varphi) = \sum_{m=0}^{(N-1)/2} \cos((2m+1)\varphi)$$

we obtained the following results.

(1) Formation of the isolated peak wave

- S_N is an “isolated peak wave” with a main peak at the center, $S_N(0) = (N+1)/2$, and zeros at both ends.

(2) Period and fundamental domain of the squared amplitude

- The squared amplitude $I_N = |S_N|^2$ has period π , and the half-wavelength interval $[-\pi/2, \pi/2]$ is its fundamental domain.

(3) Cosine sum

- By the cosine-sum formula for an arithmetic progression (Eq. (2.3)), $S_N(\varphi) = \sin((N+1)\varphi)/(2\sin\varphi)$.

(4) Normalized form near the central main peak

- With the magnified variable $u = (N+1)\varphi$, for large N the normalized amplitude is $\widehat{S}_N(\varphi) \approx \sin u/u$ and the normalized squared amplitude is $\widehat{I}_N(\varphi) \approx (\sin u/u)^2$ (Eq. (2.9)). At the same time, the horizontal width of the central main peak shrinks on the $1/(N+1)$ scale.

(5) Inverse formula for the required harmonic order

- Solve for N from a target localization half-width Δ_k and tolerance level k (numerical-substitution method, Eq. (2.15)).

All of these are elementary properties of Fourier sums, and no physical interpretation is given.

References

- [1] Teiji Takagi, *Kaiseki Gairon* (Introduction to Analysis), 3rd revised ed., Iwanami Shoten, 1961 (the chapter on Fourier series: trigonometric series, the closed form of cosine sums, the Dirichlet kernel, and the expansion of the square wave).

Appendix A: Scale estimates (numerical example; not a physical claim)

The following is merely a **numerical example** to show the reach of the numerical-substitution method **Eq. (2.15)**. It does not claim any particular physical system or physical correspondence.

Let the full width of the half-wavelength interval correspond to some large length L (regarded as a half-wavelength), and consider squeezing the central isolated peak wave to a half-width of order the Bohr radius $a_0 \approx 5.29 \times 10^{-11}$ m, used here for convenience as an example of an arbitrary small length scale (the Bohr radius is only one example of a concrete small scale and does not denote any particular physical object). Then, since the half-wavelength interval (full width π) corresponds to the physical length L , the normalized half-width is $\Delta_k = a_0/L$.

For each row, substitute $k = 0.01$ and $\Delta_k = a_0/L$ into Eq. (2.15) to compute $N_{\text{cont}}^{(\text{num})}$. To obtain the required highest odd-harmonic order as an integer, choose, per (2.15), the smallest odd integer $N_{\text{min}}^{(\text{num})}$ at least equal to $N_{\text{cont}}^{(\text{num})}$.

Numerical example (target half-width = a_0 , $k = 1\%$, computed by Eq. (2.15)):

Half-wavelength				
L	L [m]	L/a_0	$\Delta_k = a_0/L$	$N_{\text{cont}}^{(\text{num})}$
1 light-year	9.46×10^{15}	1.79×10^{26}	5.59×10^{-27}	$\sim 1.6 \times 10^{26}$

Half-wavelength L	L [m]	L/a_0	$\Delta_k = a_0/L$	$N_{\text{cont}}^{(\text{num})}$
Radius of the observable universe 2×10^{11} light-years	4.4×10^{26}	8.3×10^{36}	1.2×10^{-37}	$\sim 7.5 \times 10^{36}$
	1.89×10^{27}	3.58×10^{37}	2.79×10^{-38}	$\sim 3.2 \times 10^{37}$

For example, at $L = 2 \times 10^{11}$ light-years (merely an image of the size of the universe including unobservable regions), target half-width = Bohr radius, and $k = 1\%$, the normalized half-width is $\Delta_k = a_0/L \approx 2.79 \times 10^{-38}$, and the continuous value of Eq. (2.15) is $N_{\text{cont}}^{(\text{num})} \approx 3.2 \times 10^{37}$. The required order N is proportional to the half-wavelength L and inversely proportional to the target half-width.

To repeat, this appendix is an arithmetic application of the numerical-substitution method (Eq. (2.15)) to given scale ratios, and does not claim physical reality or any physical process.

Appendix B: Reproducing the figures

Figure 1 is generated by `figures/make_odd_harmonic_figure.py` and Figure 2 by `figures/make_odd_harmonic_scaling_f` (outputs: `figures/fig01_odd_harmonic_localization.png / .svg` and `figures/fig02_odd_harmonic_scaling.png / .svg`). Both plot, on the half-wavelength interval $\varphi \in [-\pi/2, \pi/2]$ (horizontal axis $100\% = 180^\circ$, $\pm 50\% \leftrightarrow \pm 90^\circ$), the amplitude S_N evaluated, squared, and normalized to a maximum of 1.0. Figure 2 has three panels with the horizontal axis taken as $\pm 10\%$, $\pm 1\%$, $\pm 0.1\%$ (by factors of 10) for $N = 99, 999, 9999$. Each script first verifies that, for each N , the direct sum of Eq. (2.1) and the closed form of Eq. (2.3) agree to machine precision (maximum absolute difference $\lesssim 10^{-9}$) before plotting.